

Adaptive basis set generation methods

February 24, 2024

1 Introduction

The Density-Based Basis-Set Correction (DBBSC) method [6] is an electronic structure theory method based on the one-electron density. This method introduces a functional that corrects electron correlation and allows to achieve convergence to the complete basis set limit. The high accuracy of DBCBS makes it interesting for use in solving electronic structure theory problems on quantum simulators, for future quantum computer benchmarking purposes. We study numerical tools that make the implementation of this method possible on quantum simulators with a number of available qubits up to 28. This corresponds to a number of doubly occupied molecular orbitals equal to 14.

Existing molecular orbital discretisation methods include wavelets [1] and linear combinations of atomic orbitals, for instance Gaussian-type orbitals. We focus on the latter. Existing methods for compressing molecular orbitals, such as the active space method [7], do not easily allow to reduce the number of MOs to minimal sizes while respecting chemical accuracy. Other existing methods based on the Cholesky decomposition [5] of the Hamiltonian operator employ low rank approximations to improve storage requirements and operation count. Such class of methods is based on matrix or tensor compression from a linear algebra point of view. Here, we investigate alternative density-based solutions, that achieve low computational and storage cost by approximating the electronic density function using smaller basis sets. The advantage of this approach is that the resulting smaller basis set can be used for multiple purposes. Namely, it can directly be used as a discretisation basis for minimizing any given energy functional, since it is Hamiltonian-independent.

The main motivation behind our approach is the presence of linear dependencies, which are intrinsic to any given atomic orbital basis set of Gaussian-type orbitals. Inspired by density fitting methods [4], our strategy aims at detecting linear dependencies. This technique produces a smaller basis set using a density-based criterion, which is the approximation of a given target electron density under any given norm. In the present work, we propose new adaptive basis set generation methods based on density fitting. We divide the computation time into an *offline phase*, performed on a classical computer, and an *online phase*, performed on a quantum simulator. The calculation of the target electron density, used as a reference, and the adapted basis generation are both performed during the offline phase.

The present manuscript is organized as follows. The first part describes the mathematical background of our adapted basis set generation methods and the second part contains numerical results. The performance of our method is numerically assessed on small molecules.

2 Theory

We briefly describe the theory of the DBBSC method, originally proposed in [2]. Let us consider a system of N electrons and the electronic Hamiltonian

$$H = T + W_{ee} + V_{ne},$$

where T is the kinetic operator, W_{ee} is the electron repulsion operator, given by $\sum_{i < j}^N r_{ij}^{-1}$, and V_{ne} is the nuclei-electron attraction operator. The exact electronic ground state energy problem reads

$$E_0 = \min_{\Psi \in W} \langle \Psi, H \Psi \rangle, \quad (1)$$

where the exact wave function lives in the space defined as

$$W := \{\Psi \in \wedge^N H^1(\mathbb{R}^3, \mathbb{C}) : \langle \Psi, \Psi \rangle = 1\}.$$

We discretise problem of Eq. (1) by introducing a given finite basis set, denoted by $B \subset H^1(\mathbb{R}^3, \mathbb{C})$, and the discretised wave function space

$$W^B := \{\Psi \in \wedge^N \text{Span}(B) : \langle \Psi, \Psi \rangle = 1\}.$$

The discretised ground state problem then reads

$$E_*^B = \min_{\Psi \in W^B} \langle \Psi, H \Psi \rangle. \quad (2)$$

The limitation of estimating the exact ground state energy by solving the discretised problem of Eq. (2) is that the error on the energy converges as $\mathcal{O}(|B|^{-1})$, due to the cusp of the exact electron wave function. In order to improve the error convergence, the DBBSC method proposes to approximate the ground state energy by solving the following discretised problem

$$E_0^B = \min_{\Psi \in W^B} (\langle \Psi, H \Psi \rangle + \bar{E}^B[n_\Psi]), \quad (3)$$

with $\bar{E}^B[n_\Psi]$ the basis set correction functional, we refer to the original work for its explicit definition. The one-electron density n_Ψ , for any given wave function $\Psi \in W^B$, lives in the space

$$D^B := \{n \in D : \exists \Psi \in W^B, \quad n_\Psi = n\},$$

with

$$D = \{n \in L^1(\mathbb{R}^3) : n \geq 0, \quad \int_{\mathbb{R}^3} n = N, \quad \sqrt{n} \in H^1(\mathbb{R}^3)\}.$$

The problem of Eq. (3) is solved self-consistently [3].

2.1 Slater determinants

In Hartree-Fock and post Hartree-Fock methods, a special form of wave function is considered, known as a *Slater determinant*, defined as

$$\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N) = \frac{1}{\sqrt{N!}} \det(\phi_i(\mathbf{r}_j))_{1 \leq i, j \leq N} \quad (4)$$

where each ϕ_i is a normalised one-electron function of $H^1(\mathbb{R}^3, \mathbb{C})$ referred to as Molecular Orbital (MO). It can be easily verified that Ψ in the form of Eq. (4) is indeed an element of W . For any given Slater determinant $\Psi \in W$, the one-electron density is given in terms of the Molecular Orbitals (MOs) as

$$n_\Psi = \sum_{i=1}^N |\phi_i|^2. \quad (5)$$

2.2 Approximating the density

Under the Linear Combination of Atomic Orbitals (LCAO) approximation, the discretisation basis B is denoted by $B = \{\chi_\mu\}_{1 \leq \mu \leq N_b}$, where each χ_μ is an atomic orbital basis function. Such discretisation yields an approximation of the exact MOs on the basis of AOs. The problem unknown is the coefficient matrix $\mathbf{C} \in \mathbb{R}^{N \times N_b}$ in the decomposition

$$\forall 1 \leq i \leq N, \quad \phi_i = \sum_{\mu=1}^{N_b} C_{i\mu} \chi_\mu.$$

Substituting this MO expression in the one-electron density of Eq. (5), one obtains the following expansion of the density on the AOs:

$$n_\Psi = \sum_{\mu=1}^{N_b} \sum_{\nu=1}^{N_b} D_{\mu\nu} \chi_\mu \chi_\nu, \quad (6)$$

where $\mathbf{D} = \mathbf{C}^\top \mathbf{C}$ is the so-called *density matrix*, of dimension $N_b \times N_b$.

2.3 Problem formulation

The problem consists in constructing a subset $I \subseteq \{1, \dots, N_b\}$ of any given size M such that if the restricted basis $B_I := \{\chi_\mu\}_{\mu \in I}$ is used to discretise the problem of Eq. (3), i.e. if the minimisation space is W^{B_I} instead of W^B , then the accuracy loss on the estimated ground state energy with respect to using the full basis B , i.e. the error

$$|E_0^B - E_0^{B_I}|,$$

remains small.

Our main idea for solving this problem can be described as follows. Let us denote by $\Psi_0 \in W^B$ the minimiser of problem of Eq. (3) and by $n_{\Psi_0} \in D^B$ the associated electron density. Given a target size M , we seek the optimal subset of B of the form B_I with $|I| = M$, obtained by solving the following optimisation problem

$$\min_{\substack{I \subseteq \{1, \dots, N_b\} \\ |I|=M}} \min_{m \in D^{B_I}} \|n_{\Psi_0} - m\|,$$

where $\|\cdot\|$ is any given norm. Note that the optimisation criterion we used is the orthogonal projection error (also known as best approximation error) with respect to the reference electron density. In other words, the problem boils down to selecting the index subset of size M such that the best approximation error is minimised.

Due to the expression of the density given by Eq. (6), any function $m \in D^{B_I}$ is of the form

$$m = \sum_{\mu \in I} \sum_{\nu \in I} C_{\mu\nu} \chi_\mu \chi_\nu.$$

2.4 Algorithms

To summarize, the main ingredients given as an input to our algorithm is

- given molecular geometry,
- atomic orbital basis set,
- a starting density matrix.

We studied three different choices of density matrices: Hartree-Fock (HF), Configuration Interaction Singles and Doubles (CISD) and Configuration Interaction Perturbatively Selected Iteratively (CIPSI).

2.4.1 Assembly of the Gram matrix

The first step is the assembly of the Gram matrix (i.e. the overlap matrix) of the orbital products, given a user-defined inner product $\langle \cdot, \cdot \rangle$. Let us denote by $\mathbf{T}$ the tensor of dimension $N_b \times N_b \times N_b \times N_b$, with entries defined as

$$T_{\mu\nu\kappa\lambda} = D_{\mu\nu} D_{\kappa\lambda} \langle \chi_\mu \chi_\nu, \chi_\kappa \chi_\lambda \rangle.$$

The Gram matrix $\mathbf{G}$, of dimension $N_b^2 \times N_b^2$ is then formed by folding pairwise the first two and the last two dimensions of the tensor $\mathbf{T}$. Then we also regroup by orbital types. Another option for initialising the Gram matrix consists in $\mathbf{R}^\top \mathbf{G} \mathbf{R}$, where $\mathbf{R}$ is the atomic restriction matrix that discards orbital products centered on midbonds.

2.4.2 Selecting orbital products

We seek a selection of orbital products of target size $M_* \ll N_b^2$ that best approximate the one-electron density. This selection can be performed using a criterion based on minimizing the orthogonal projection error with respect to the full space spanned by orbital products, in which the density belongs to. Orbital products are discarded using a pivoted Cholesky decomposition performed on the Gram matrix $\mathbf{G}$, also known as Gram-Schmidt procedure with pivoting. The orthonormalization procedure terminates when the orthogonal projection error is smaller than a given threshold. Note that the entries of the Gram matrix are weighted by the density matrix. Therefore, at the first iteration of the pivoted Cholesky decomposition, the selected pivot is influenced by the value of the density matrix.

The pivoted Cholesky decomposition under a given tolerance value $\varepsilon > 0$, performed on the matrix $\mathbf{G}$, yields an approximate rank $M \leq N_b$ of $\mathbf{G}$ and a selection of M rows (and columns) that approximately span the full row space under the tolerance value. We propose to use the atomic orbitals corresponding to the selected indices to build the set of adaptive basis functions. This strategy leads to our algorithm named ABS-0, for Adaptive Basis Set generation.

Additionally, it may be of interest to impose maximal orbital type constraints to the adaptive basis set, in favour of computational efficiency within the integral evaluation procedure. The preliminary treatment of the Gram matrix $\mathbf{G}$ that enables this constraint can be presented as follows. The main idea is to construct an intermediary atomic orbital basis respecting orbital type constraints. We propose two possible methods. For each $1 \leq \mu \leq N_b$, if the atomic orbital χ_μ has orbital type strictly higher than d -type, then

- (a) ABS-1: discard χ_μ .
- (b) ABS-2: for each orbital type $t \in \{s, p, d\}$, create a new atomic orbital having the orbital exponent and atomic center of χ_μ , but orbital type t .

It is assumed that all atomic basis functions are normalized, so that the diagonal of $\mathbf{G}$ consists of one entries.

3 Numerical experiments

3.1 Implementation details

The quantum simulator uses the Hamiltonian in the second quantisation formulation, that reads

$$\hat{H} = \sum_{\sigma} \sum_{ij}^{N_{\text{MO}}} h_{ij} \hat{a}_{i\sigma}^\dagger \hat{a}_{j\sigma} + \sum_{ijkl}^{N_{\text{MO}}} h_{ijkl} \hat{a}_{i\sigma}^\dagger \hat{a}_{k\sigma}^\dagger \hat{a}_{j\sigma} \hat{a}_{l\sigma}, \quad (7)$$

where N_{MO} is the number of basis functions used to discretise the molecular orbitals. Note that our method allows to reduce the number of MOs in this expression.

3.2 Results

The first experiment allows to compare the results obtained from a DFT/B3LYP calculation on the water molecule at equilibrium geometry, using state-of-the-art auxiliary basis sets (RI and JKFIT) and our adaptive basis sets, obtained with methods ABS- x for $x = 0, 1, 2$. The aim of this comparison is to study which method achieves better accuracy with less basis functions. The approximation error is measured on ground state energies using the full atomic orbital basis set as a reference calculation. We are also interested in assessing the convergence of the approximation error of our method for varying tolerance values in the Cholesky decomposition.

4 Perspectives

Our method can be combined with orbital exponent and contraction coefficient optimization methods, namely with the one proposed in [7]. A possible direction that allows to combine both methods is to first apply our method for basis set size reduction, then apply their method on our resulting adapted basis set for further optimization, in the goal of increasing the accuracy using a fixed number of MOs.

References

- [1] Luigi Genovese and Thierry Deutsch. Flexibilities of wavelets as a computational basis set for large-scale electronic structure calculations. In *Density Functional Theory: Modeling, Mathematical Analysis, Computational Methods, and Applications*, pages 479–523. Springer, 2022.
- [2] Emmanuel Giner, Barthélemy Pradines, Anthony Ferté, Roland Assaraf, Andreas Savin, and Julien Toulouse. Curing basis-set convergence of wave-function theory using density-functional theory: A systematically improvable approach. *The Journal of chemical physics*, 149(19), 2018.
- [3] Emmanuel Giner, Diata Traore, Barthélemy Pradines, and Julien Toulouse. Self-consistent density-based basis-set correction: How much do we lower total energies and improve dipole moments? *The Journal of Chemical Physics*, 155(4), 2021.
- [4] Andreas Hesselmann, Emmanuel Giner, Peter Reinhardt, Peter J Knowles, Hans-Joachim Werner, and Julien Toulouse. A density-fitting implementation of the density-based basis-set correction method. *arXiv preprint arXiv:2309.10390*, 2023.
- [5] Felix Hummel, Theodoros Tsatsoulis, and Andreas Grüneis. Low rank factorization of the coulomb integrals for periodic coupled cluster theory. *The Journal of chemical physics*, 146(12), 2017.
- [6] Diata Traore. *L’extension de la méthode de correction de base par la densité pour les calculs de propriétés moléculaires*. Theses, Sorbonne Université, July 2023.
- [7] Weishi Wang and James D Whitfield. Basis set generation and optimization in the nisq era with quiqbox. jl. *Journal of Chemical Theory and Computation*, 19(22):8032–8052, 2023.